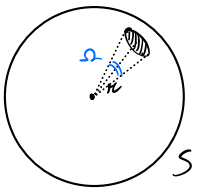
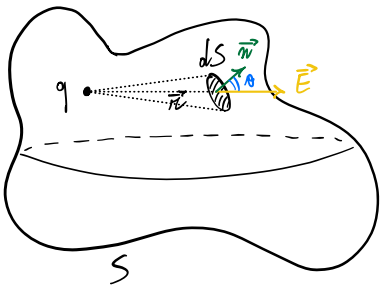
Angolo solido



$$\Omega = \frac{S}{r^2}$$

$$\Omega_{max} = \frac{4\pi r^2}{r^2} = 4\pi$$

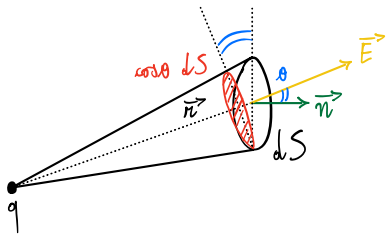
Vediamo se ciò vale per altre superfici:



$$d\Phi = \vec{E}(r)\cdot\vec{n}\,dS = \frac{q}{4\pi\varepsilon_0 r^2}\,\vec{u}_r\cdot\vec{n}\,dS = \frac{q}{4\pi\varepsilon_0}\cdot\frac{\cos\theta}{r^2}\,dS$$

$$= \frac{q}{4\pi\varepsilon_0}\,d\Omega \Rightarrow \text{angolo solido}$$

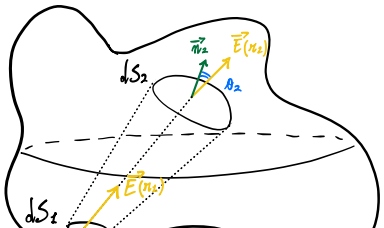
Quindi:



$$\Phi_S(\vec{E}) = \int d\Phi = \int_0^{4\pi} \frac{q}{4\pi\varepsilon_0}\,d\Omega = \frac{q}{4\pi\varepsilon_0}\,4\pi = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \Phi_S(\vec{E}) = \frac{q}{\varepsilon_0}$$ ∀ superficie, ∀ posizione interna della carica

E se ci fosse una carica esterna?



$$d\Phi = -E(r_1)\cos\theta_1\,dS_1 + E(r_2)\cos\theta_2\,dS_2$$

$$= -\frac{q}{4\pi\varepsilon_0}\underbrace{\frac{\cos\theta_1\,dS_1}{r_1^2}}_{d\Omega} + \frac{q}{4\pi\varepsilon_0}\underbrace{\frac{\cos\theta_2\,dS_2}{r_2^2}}_{d\Omega} = 0$$



TEOREMA DI GAUSS

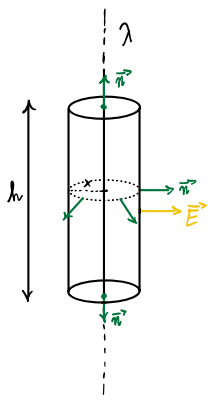
$$\Phi_S(\vec{E}) = \oint_S \vec{E}\cdot\vec{n}\,dS = \frac{q_{int}}{\varepsilon_0}$$

- q_{int}? $\Rightarrow$ $q_{int} = \sum_{i=1}^{n} q_i$

  $\Rightarrow$ $q_{int} = \int_\tau dq = \int_\tau \rho\, d\tau$ } Nel caso di carica non puntiforme

- Il fatto che Φ dipenda solo dalle sorgenti racchiuse nella superficie vale $\forall\ \vec{v}$ campo vettoriale t.c. $\vec{v} \propto \frac{1}{r^2}$
  (esempio: campo gravitazionale)

- Esempio: filo infinito



$$d\Phi_{base} = \vec{E}\cdot\vec{n}\,dS = 0$$

$$d\Phi_{lat} = \vec{E}\cdot\vec{n}\,dS = E(x)\,dS$$

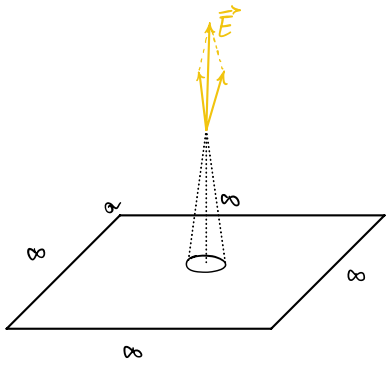
$$\Phi_{lat}(\vec{E}) = \int_{lat} E(x)\,dS = E(x)\int_{lat} dS = E(x)\,2\pi x h = \frac{q_{int}}{\varepsilon_0} = \frac{\lambda h}{\varepsilon_0}$$

$\frac{q_{int}}{\varepsilon_0}$: per Gauss; $\frac{\lambda h}{\varepsilon_0}$: $\lambda = \frac{q_{int}}{h}$

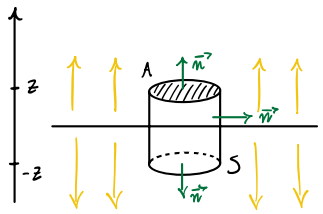
$$\Rightarrow E(x)\,2\pi x h = \frac{\lambda h}{\varepsilon_0}$$

$$\vec{E}(x) = \frac{\lambda}{2\pi\varepsilon_0 x}\vec{u}_r \quad \Rightarrow \text{risultato già trovato!}$$

• Esempio: piano infinito



Anche in questo caso le componenti tangenziali al piano si annullano



$$d\Phi_{lat} = 0$$

$$d\Phi_{base} = 2\, d\Phi_A = d\Phi_S$$

$$\Phi_S(\vec{E}) = 2\int_A \vec{E}\cdot\vec{n}\, dS = 2\int_A E(z)\, dS = 2E(z)\int_A dS = 2E(z)A$$

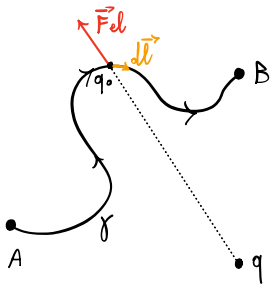
$$\Rightarrow 2E(z)A = \frac{q_{int}}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} \quad \Rightarrow \quad E(z) = \frac{\sigma}{2\varepsilon_0} \quad \Rightarrow \text{non dipende da } z!!$$

$\frac{q_{int}}{\varepsilon_0}$ → Gauss; $\frac{\sigma A}{\varepsilon_0}$ → $\sigma = \frac{q_{int}}{A}$

$$\vec{E} = \begin{cases} \frac{\sigma}{2\varepsilon_0}\vec{u}_z & z > 0 \\ -\frac{\sigma}{2\varepsilon_0}\vec{u}_z & z < 0 \end{cases}$$

Campo elettrico di un piano infinito

ENERGIA POTENZIALE ELETTROSTATICA E POTENZIALE



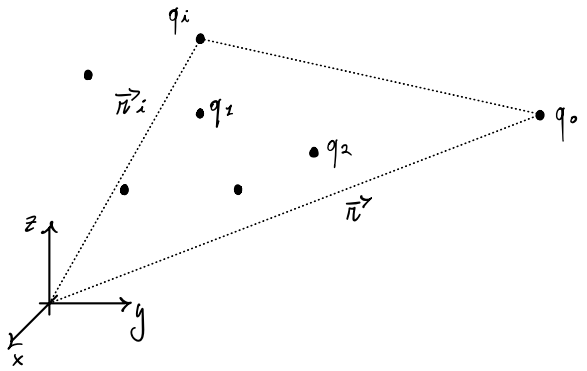
$$\mathcal{L}_{el} = \int_A^B \vec{F}el\cdot d\vec{l} = \int_A^B \frac{qq_0}{4\pi\varepsilon_0 r^2}\vec{u}_r\cdot d\vec{l} = \int_A^B \frac{qq_0}{4\pi\varepsilon_0 r^2}\, dr =$$

$$= \frac{qq_0}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right) = -\left(\frac{qq_0}{4\pi\varepsilon_0 r_B} - \frac{qq_0}{4\pi\varepsilon_0 r_B}\right) = -\Delta E_p(r)$$

$$\boxed{\mathcal{L}^{el} = -\Delta E_p}$$

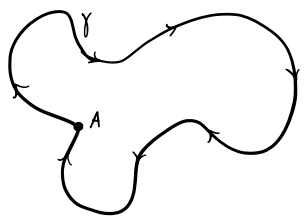
$$\Rightarrow E_p(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_0 q}{r} + c \qquad (c = 0 \text{ per convenzione})$$

Consideriamo:



$$E_p(r) = \frac{q_0}{4\pi\varepsilon_0} \sum_{i=1}^{n} \frac{q_i}{|\vec{r} - \vec{r}_i|}$$

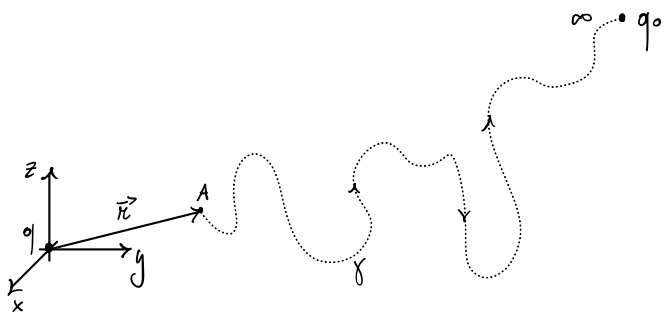
Consideriamo:



$$\mathcal{L}^{\gamma}_{AA} = \oint_{\gamma} \vec{F}_{el} \cdot d\vec{l} = -\Delta E_p \overset{\text{perché } \vec{F}_{el} \text{ è conservativa}}{=} 0$$

$$\Rightarrow \boxed{\oint_{\gamma} \vec{E} \cdot d\vec{l} = 0}$$

OSS. Cosa intendiamo con energia potenziale elettrostatica in questo caso?



Porto q_0 a distanza infinita da q

$$\Delta E_p = E_p(r) - \underbrace{E_p(\infty)}_{=0} = \frac{1}{4\pi\varepsilon_0} \frac{qq_0}{r}$$

$$\Rightarrow \mathcal{L}_{el} = -\Delta E_p \overset{(>0)}{}$$

nel caso di 2 cariche positive

$\Downarrow$

$\vec{E}$ si oppone al moto

$$\Rightarrow \mathcal{L}_{ext} = -\mathcal{L}_{el}$$

$$\boxed{V = \frac{E_p}{q_0}}$$ Potenziale elettrostatico $\left[\frac{J}{C}\right] = [V]$

• Per cariche puntiformi

$$E_p = \frac{1}{4\pi\varepsilon_0}\frac{q q_0}{r} \quad \Longrightarrow \quad V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

• Per n - cariche puntiformi

$$E_p = \frac{1}{4\pi\varepsilon_0} q_0 \sum_{i=1}^{n} \frac{q_i}{|\vec{r}-\vec{r}_i|} \quad \Longrightarrow \quad V = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \frac{q_i}{|\vec{r}-\vec{r}_i|}$$

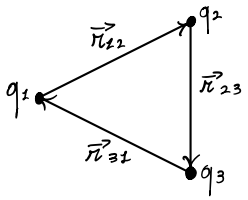
• Per cariche continue

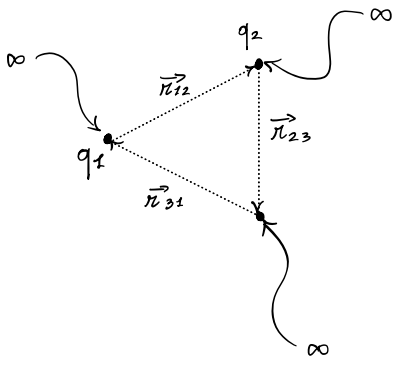
$$E_p = \frac{1}{4\pi\varepsilon_0} q_0 \int \frac{dq}{|\vec{r}-\vec{r}'|} \quad \Longrightarrow \quad V = \frac{1}{4\pi\varepsilon_0} \int \frac{dq}{|\vec{r}-\vec{r}'|} = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(r)\, d\tau}{|\vec{r}-\vec{r}'|}$$

$$\Longrightarrow \mathcal{L}_{el} = -\Delta E_p = -q_0 \Delta V \quad \Longrightarrow \quad \boxed{\mathcal{L}_{el} = -q_0 \Delta V}$$, dove ΔV = differenza di potenziale

$$\Longrightarrow \int_A^B \frac{\vec{F}_{el}}{q_0} \cdot d\vec{l} = -\frac{\Delta E_p}{q_0} \quad \Longrightarrow \quad \boxed{\int_A^B \vec{E} \cdot d\vec{l} = -\Delta V} = -\int_{V_A}^{V_B} dV \quad \Longrightarrow \quad \boxed{dV = -\vec{E} \cdot d\vec{l}}$$

• Esempio Notevole





$$\mathcal{L}_1 = 0$$

$$\mathcal{L}_2 = -\frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$$

$$\mathcal{L}_3 = -\frac{q_1 q_3}{4\pi\varepsilon_0 r_{31}} - \frac{q_2 q_3}{4\pi\varepsilon_0 r_{23}}$$

$$\mathcal{L}_{ext} = -\mathcal{L}_1 - \mathcal{L}_2 - \mathcal{L}_3 = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} + \frac{q_2 q_3}{4\pi\varepsilon_0 r_{23}} + \frac{q_3 q_1}{4\pi\varepsilon_0 r_{31}} = \frac{1}{2}\sum_{i=1}^{n} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} = \frac{1}{2}\sum_{i=1}^{n} E_{p\,ij}$$

$$\mathcal{L}_{ext} = \frac{1}{2} \sum_{i=1}^{n} q_i \sum_{j \neq i}^{m} \frac{q_j}{4\pi\varepsilon_0 r_{ij}} = \frac{1}{2} \sum_{i=1}^{n} q_i V_i$$

dove V_i = potenziale che "sente" la carica i

$$\Rightarrow \boxed{\mathcal{L}_{ext} = \frac{1}{2} \sum_{i=1}^{n} q_i V_i}$$

$$\Rightarrow \boxed{\mathcal{L}_{ext} = \frac{1}{2} \int d\mathcal{V}\, q\, V}$$